**Quantitative Macro Theory** **HW #4**
**Spring 2018** **Due at 5pm on *Friday* 2/16/2018**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (100 points) **Income-Fluctuation problem**

   Consider the problem

$$V(a, y_i) = \max_{c,a'} u(c) + \rho\beta \sum_{j=1}^{3} \Gamma_{i,j} V(a', y_j)$$

   subject to

$$c + a' = Ra + y_i,\ a' \geq \underline{a}.$$

   Calibration:

   - The probability of survival is $\rho = 0.975$.
   - $u(c) = \frac{c^{1-\sigma}}{(1-\sigma)}$, with $\sigma = 2$.
   - The discount rate, $\beta = 0.92$.
   - Gross interest rate, $R = 1.02$.
   - Income is given by $y$. Use a three-state Markov chain $\Gamma$ that we loosely identify with "super-rich" ($y_1 = 16,000$), "white-collar" ($y_2 = 80$), and "blue-collar" ($y_3 = 15$).
   - The first state provides the opportunity and incentive for a high concentration of earnings and wealth (see Castañeda, Diaz-Gimenez, and Rios-Rull, 2003). $\Gamma_{2,1} = 0.001$, $\Gamma_{1,1} = 0.02$.
   - The persistence of the latter two states ensures that earnings display a sizable positive autocorrelation: $\Gamma_{2,2} = 0.78$, $\Gamma_{3,3} = 0.964$.
   - This specification requires 6 parameters in the Markov transition matrix but we reduce it to 4 by setting the probability of moving from blue-collar to super-rich and vice-versa to zero. $\Gamma_{3,1} = 0$, $\Gamma_{1,3} = 0$, $\Gamma_{1,2} = 0.98$, $\Gamma_{2,3} = 0.219$, $\Gamma_{3,2} = 0.036$.
   - $\underline{a} = 0$.

   (a) (20 points) Solve this problem using discretization and policy function iteration with 200 and 1000 points in the grid for $a$ with the same range. Report how much time each case takes. Plot $a'(a, y_2)$ for the two cases in the same figure to compare the results.

   (b) (20 points) Plot $a'(a, y_1)$ for the case of 1000 grid points (include a 45 degree line). Then, plot $a'(a, y_3)$ for the same case (include a 45 degree line, too). What explain the difference in the saving behavior of "super-rich" and "blue-collar" workers?

(c) (20 points) Now solve the problem using 50 points in the grid and Chebyshev interpolation using polynomial of order N=5. Report how much time it takes. Repeat the plots in part (b) with this new solution and the solution using discretization with 1000 points.

(d) (20 points) Now solve the problem using 50 points in the grid and the endogenous grid point method. Report how much time it takes. Repeat the plots in part (b) with this new solution and the solution using discretization with 1000 points.

(e) (20 points) Pick the method it worked better for you. What is it? Solve the problem using that method but now setting $\underline{a} = -50$. Plot $a'(a, y_3)$ for this case and the case with $\underline{a} = 0$ in the same figure. Explain how and why the saving behavior of "blue-collar" workers change.